

Certified Polyhedral Projection for Counterfactual Explanations

Generating Valid, Proximal, and Robust Counterfactuals
via Preimage Approximation

Gabriele Pintos

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What Are Counterfactual Explanations?

Given a classifier $f : \mathbb{R}^d \rightarrow \mathbb{R}^L$ and a query $\mathbf{x}_0$ classified as class l :

Goal

Find $\mathbf{x}' \in \mathbb{R}^d$ that is **classified as target class** $t \neq l$ and is **as close as possible** to $\mathbf{x}_0$.

Example (MNIST): “What minimal change turns a **digit 1** into a **digit 7**?”

Desirable properties:

- ① **Validity** – guaranteed to be classified as the target class
- ② **Proximity** – minimal distance to the original input
- ③ **Plausibility** – stays close to the data manifold
- ④ **Robustness** – remains valid under small perturbations

Limitations of Existing Approaches

Standard gradient-based methods solve:

$$\mathbf{x}' = \arg \min_{\mathbf{z}} \|\mathbf{x}_0 - \mathbf{z}\|_p + \lambda \cdot \mathcal{L}_{\text{target}}(f(\mathbf{z}))$$

Problems

- **Non-convex landscape** $\Rightarrow$ local minima, no global optimality
- **No validity guarantee** $\Rightarrow$ counterfactuals may land near the decision boundary
- **No robustness certificate** $\Rightarrow$ tiny perturbations can flip the prediction
- **Out-of-distribution** $\Rightarrow$ adversarial artifacts, not meaningful changes

Our approach: replace gradient descent on a non-convex loss with **convex projection onto certified polytopes**.

Core Idea: From Non-Convex Search to Convex Projection

Key Insight

If we can decompose the target class preimage into a **union of convex polytopes** $\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_N$ where the classifier is *guaranteed* to predict class t , then finding the counterfactual reduces to:

$$\mathbf{x}' = \text{proj}_{\bigcup_i \mathcal{P}_i}(\mathbf{x}_0) = \arg \min_{\mathbf{z} \in \bigcup_i \mathcal{P}_i} \|\mathbf{x}_0 - \mathbf{z}\|_2$$

- Each individual projection $\text{proj}_{\mathcal{P}_i}(\mathbf{x}_0)$ is a **convex QP** $\Rightarrow$ **global optimum**
- The overall nearest projection is found by comparing across all $\mathcal{P}_i$
- **No gradient descent**, no local minima, no hyperparameter λ

Two-Phase Architecture

Phase I: Atlas Construction (Offline)

For each training sample $\mathbf{c}_i$ of class t :

- 1 Define L_p ball $\mathcal{B}_p(\mathbf{c}_i, \varepsilon)$
- 2 Run **LiRPA** backward-mode bound propagation
- 3 Obtain linear bounds $A_i, \mathbf{b}_i$ such that:

$$A_i \mathbf{x} + \mathbf{b}_i \leq f_t(\mathbf{x}) - f_j(\mathbf{x}) \quad \forall j \neq t$$

- 4 Store certified polytope:

$$\mathcal{P}_i = \{\mathbf{x} : A_i \mathbf{x} + \mathbf{b}_i \geq \mathbf{0}\} \cap \mathcal{B}_p(\mathbf{c}_i, \varepsilon)$$

$\Rightarrow$ Every point in $\mathcal{P}_i$ is guaranteed class t

Phase II: Counterfactual Query (Online)

Given query $\mathbf{x}_0$:

- 1 Use **BVH** spatial index to prune candidates
- 2 For each candidate polytope $\mathcal{P}_i$, solve QP:

$$\min_{\mathbf{z}} \|\mathbf{z} - \mathbf{x}_0\|_2^2$$

subject to:

$$A_i \mathbf{z} + \mathbf{b}_i \geq \mathbf{0}$$

$$\|\mathbf{z} - \mathbf{c}_i\|_p \leq \varepsilon$$

- 3 Return projection with **minimum distance**

$\Rightarrow$ Globally optimal among all polytopes visited

LiRPA: Linear Relaxation based Perturbation Analysis

Goal: For a neural network f and an input region $\mathcal{B}_p(\mathbf{c}, \varepsilon)$, find linear functions that **bound** the network output from below.

One-vs-All Formulation

We wrap the classifier with a constraint layer:

$$g_j(\mathbf{x}) = z_t(\mathbf{x}) - z_j(\mathbf{x}) \quad \text{for all } j \neq t$$

where $z_k(\mathbf{x})$ is the logit for class k . The sample is classified as t iff $g_j(\mathbf{x}) \geq 0 \forall j$.

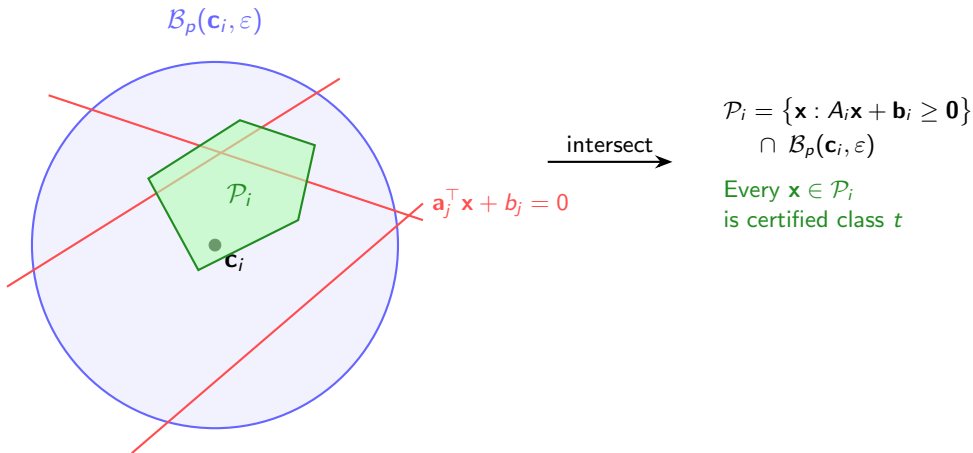
LiRPA computes **linear lower bounds**:

$$\mathbf{a}_{ij}^\top \mathbf{x} + b_{ij} \leq g_j(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{B}_p(\mathbf{c}, \varepsilon)$$

Therefore: $\mathbf{a}_{ij}^\top \mathbf{x} + b_{ij} \geq 0 \implies g_j(\mathbf{x}) \geq 0 \implies \text{class } t \text{ certified.}$

Stacking all $K-1$ constraint rows gives the matrix $A_i \in \mathbb{R}^{(K-1) \times d}$ and vector $\mathbf{b}_i \in \mathbb{R}^{K-1}$.

From Bounds to Certified Polytopes



The **Certified Atlas** is the collection $\{\mathcal{P}_i\}_{i=1}^{N_t}$ for all training samples of class t .

BVH Spatial Indexing

Problem: With N polytopes per class, naïve search requires N QP solves per query. In 784 dimensions (MNIST), each QP costs “a lot” $\Rightarrow$ **intractable**.

Bounding Volume Hierarchy (BVH)

- Build a **binary tree** over polytope bounding boxes
- Split along axis of maximum spread (median split)
- Each internal node stores the AABB of its children
- **Branch-and-bound** search: prune subtrees whose bounding box is farther than current best

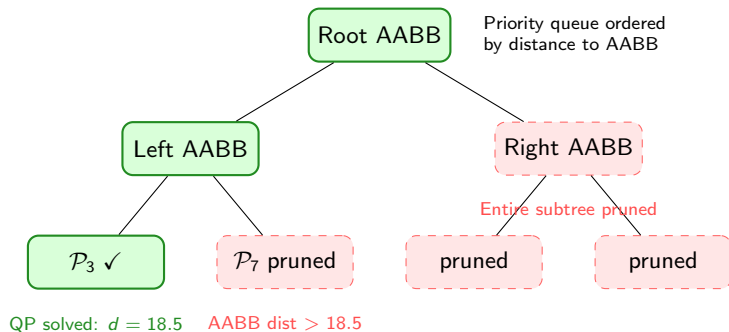
Complexity:

- Construction: $O(N \log N)$
- Query: $O(\log N)$ QP solves
- Preserves **exact** optimality

Empirical results (MNIST):

- ~ 1000 polytopes per class
- BVH: $\sim 2\text{--}5$ QPs per query
- vs. exhaustive: 1000 QPs
- $200\text{--}500\times$ **speedup**

BVH Branch-and-Bound Search



Key: The priority queue ensures we always explore the most promising subtree first. Once we find a good projection, most of the tree is pruned away.

Certified Robustness via Polytope Erosion

Goal: The counterfactual $\mathbf{x}'$ should remain valid even under adversarial perturbation of radius δ .

δ -Robust Counterfactual

We want the **entire ball** $\mathcal{B}_q(\mathbf{x}', \delta)$ to lie inside the certified polytope $\mathcal{P}_i$:

$$\mathcal{B}_q(\mathbf{x}', \delta) \subseteq \mathcal{P}_i \implies \text{any perturbation in } \mathcal{B}_q(\mathbf{x}', \delta) \text{ stays class } t$$

This is achieved by **eroding** the constraints inward:

$$\text{Halfspace erosion: } \mathbf{a}_j^\top \mathbf{x} + b_j \geq \delta \cdot \|\mathbf{a}_j\|_{q^*} \quad \left(\frac{1}{q} + \frac{1}{q^*} = 1 \right)$$

$$\text{Ball shrinkage: } \|\mathbf{x} - \mathbf{c}_i\|_p \leq \varepsilon - \delta \cdot \mathcal{C}(q \rightarrow p) \quad \left(\mathcal{C}(q \rightarrow p) = d^{\max(0, 1/p - 1/q)} \right)$$

$$\text{Box shrinkage: } \|\mathbf{x} - \mathbf{c}_i\|_\infty \leq \varepsilon - \delta$$

Setting $\delta = 0$ recovers the standard (non-robust) formulation.

The robustness norm q can **differ** from the certification norm p (cross-norm robustness).

Cross-Norm Robustness: Blending Different L_p Norms

The certification norm p (atlas) and the robustness norm q (threat model) can **differ**. Two mechanisms handle this:

1. Halfspace erosion via the **dual norm** q^*

To ensure $\mathcal{B}_q(\mathbf{x}', \delta)$ satisfies $\mathbf{a}_j^\top \mathbf{x} + b_j \geq 0$, we need: $\mathbf{a}_j^\top \mathbf{x}' + b_j \geq \delta \cdot \|\mathbf{a}_j\|_{q^*}$ where $\frac{1}{q} + \frac{1}{q^*} = 1$.

Why? Hölder's inequality: $|\mathbf{a}_j^\top \Delta \mathbf{x}| \leq \|\mathbf{a}_j\|_{q^*} \|\Delta \mathbf{x}\|_q \leq \delta \|\mathbf{a}_j\|_{q^*}$.

Robustness norm q	1	2	∞
Dual q^*	∞	2	1
Erosion per row	$\max_k a_{jk} $	$\ \mathbf{a}_j\ _2$	$\sum_k a_{jk} $

2. Ball shrinkage via **norm embedding**

For $p_2 > p_1$: $\|\mathbf{x}\|_{p_2} \leq \|\mathbf{x}\|_{p_1}$ (free direction), but $\|\mathbf{x}\|_{p_1} \leq d^{1/p_1 - 1/p_2} \|\mathbf{x}\|_{p_2}$ (costly).

If $\Delta \mathbf{x} \in \mathcal{B}_q(\mathbf{x}', \delta)$, we must fit it inside $\mathcal{B}_p(\mathbf{c}_i, \varepsilon)$. Conversion factor: $\mathcal{C}(q \rightarrow p) = d^{\max(0, 1/p - 1/q)}$.

Norm Ordering and Its Consequences

Norm Monotonicity

For any $\mathbf{x} \in \mathbb{R}^d$ and $1 \leq p_1 < p_2 \leq \infty$:

$$\|\mathbf{x}\|_{p_2} \leq \|\mathbf{x}\|_{p_1} \leq d^{1/p_1 - 1/p_2} \|\mathbf{x}\|_{p_2}$$

Consequence for cross-norm robustness:

Case $q < p$ (e.g., L_2 robustness on L_∞ atlas):

- $\mathcal{B}_q(\mathbf{x}', \delta) \subseteq \mathcal{B}_p(\mathbf{x}', \delta)$ for free (left inequality)
- Conversion factor $\mathcal{C}(q \rightarrow p) = 1$
- The ball constraint barely shrinks

$\Rightarrow$ **Cheap** – robustness nearly free

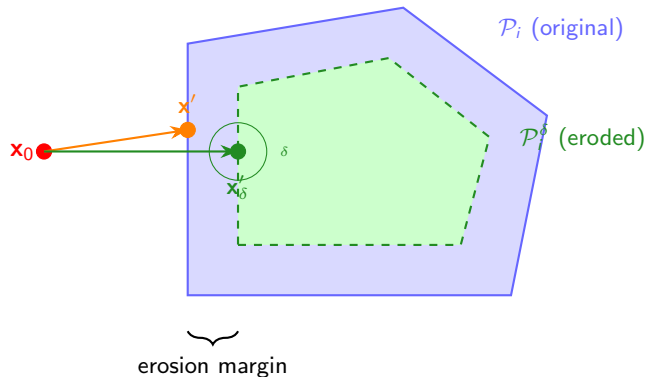
Case $q > p$ (e.g., L_∞ robustness on L_2 atlas):

- Need right inequality: $\|\Delta \mathbf{x}\|_p \leq d^{1/p-1/q} \delta$
- Conversion factor $\mathcal{C}(q \rightarrow p) = d^{1/p-1/q} \gg 1$
- The ball shrinks **dramatically** in high d

$\Rightarrow$ **Expensive** – rapid feasibility loss

Example (MNIST, $d = 784$): L_2 robustness on L_∞ atlas ($q < p$) has $\mathcal{C}(2 \rightarrow \infty) = 1$ (cheap), but L_2 robustness on L_1 atlas ($q > p$) has $\mathcal{C}(2 \rightarrow 1) = 784^{1/1-1/2} = 28$ ($28\times$ shrinkage).

Effect of δ on Counterfactuals



- Larger $\delta \Rightarrow$ smaller eroded polytope $\Rightarrow$ farther counterfactual
- If δ is too large, the eroded polytope may become **empty** (infeasible)
- Trade-off: **proximity vs. robustness**

Experimental Setup

2D Spiral Dataset:

- 5 classes, 80 samples/class (test set)
- SimpleClassifier: 2 hidden layers (64, 32)
- $\varepsilon = 0.1$, L_2 norm
- Polytopes directly visualizable

MNIST Dataset:

- 10 classes, 1000 samples/class
- CNN: 4 Conv2d + Linear (99.2% accuracy)
- $\varepsilon = 0.01$, L_∞ norm
- 784-dimensional input space

Evaluation metrics:

- ℓ_2 **distance** (proximity)
- ℓ_1 **distance** / **sparsity** (interpretability)
- **LOF score** (plausibility – closeness to target manifold)
- **QP solves** (computational cost)
- **Validity rate** (must be 100% by construction)

Solvers:

- L_∞ : SLSQP (all-linear constraints)
- L_2/L_1 : CVXPY + CLARABEL (SOCP)

Results: 2D Spiral

Atlas construction:

- 400 polytopes across 5 classes
- Feasibility: 48.8%–90.0% per class ($\varepsilon = 0.1$, L_2)
- Direct visualization of certified regions possible

Counterfactual generation:

- 100% validity rate (all counterfactuals certified)
- δ -robust counterfactuals with $\delta = 0.07$, L_2 robustness norm
- BVH: **11.6 $\times$ fewer QPs** compared to exhaustive search (6.9 vs. 80 avg)

Observations:

- Counterfactuals follow the spiral geometry of the data
- Positive margin surplus (min margin $\gg 0$) confirms deep placement inside polytopes
- Cross-class distance varies naturally with class separation

Results: MNIST

Atlas construction:

- 10,000 polytopes across 10 digit classes
- $\varepsilon = 0.01$, L_∞ norm $\Rightarrow$ 100% feasible polytopes
- BVH tree depth: 11–12 per class

Counterfactual generation (digit 1 $\rightarrow$ all other digits):

- **100% success rate**, 100% validity (certified by construction)
- Average QP solves with BVH: **1.0–2.7** per query (vs. ~ 1000 exhaustive)
- Closest target: digit 7 (avg $\ell_2 = 14.98$), farthest: digit 0 (avg $\ell_2 = 26.20$)

δ -Robustness (digit 3 $\rightarrow$ digit 8):

- L_∞ robustness: stable up to $\delta = 0.1$ (small distance increase)
- L_2 robustness: feasible up to $\delta = 0.005$ (stricter in 784D due to norm conversion)

Quality Metrics (MNIST, 180 Counterfactuals)

Metric	Mean	Median
ℓ_2 distance	23.4	22.8
ℓ_∞ distance	3.22	3.22
ℓ_1 distance	322.8	307.0
Pixels changed (of 784)	224	221
Sparsity (% pixels changed)	28.6%	28.2%
LOF score	1.55	1.29
QP solves (BVH)	3.4	2.0

- **LOF ≈ 1** indicates counterfactuals are plausible (close to inliers in target class)
- **$\sim 29\%$ of pixels changed** – counterfactuals are moderately sparse
- **BVH reduces QP solves by $\sim 300\times$** compared to exhaustive search

Summary

Contributions

- 1 **Certified Polyhedral Projection (CPP)** – transforms CE generation from non-convex optimization into a series of convex QP projections
- 2 **Provable validity** – counterfactuals lie inside LiRPA-certified polytopes $\Rightarrow$ guaranteed target-class prediction
- 3 **δ -robust counterfactuals** – polytope erosion ensures certified robustness to adversarial perturbations, with support for cross-norm robustness
- 4 **BVH spatial indexing** – enables $O(\log N)$ search instead of $O(N)$, achieving 200–500 $\times$ speedup on MNIST

Key Properties

- **No hyperparameter λ** – proximity is handled by projection, validity by certification
- **Global optimality per polytope** – convex QP guarantees
- **Offline/online separation** – atlas built once, queries are fast

Thank You

Questions?

`github.com/gabrielepintus/PreimageCounterfactualSampling`